# Transported Graph Hessian Trend Filtering

`gflow` graph trend filtering project notes

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**Abstract**

This note develops a graph trend-filtering formulation based on directional fibers and a graph connection. The starting point is the oriented incidence operator, viewed not only as an edge-difference matrix but as a collection of directional derivatives attached to each vertex. To define a graph Hessian, one must compare directional derivatives at neighboring vertices. That comparison is not well-defined from edge lengths alone: a direction at one vertex must be matched or transported to a direction at the neighboring vertex. We therefore separate the construction into two pieces: a first-difference fiber $\mathcal{F}_u(f)$, and a transport rule $T_{u \to u'}$ between neighboring fibers. The resulting transported second difference is a graph analogue of a Hessian, and its $\ell_1$ norm gives a locally adaptive trend-filtering penalty.

## 1 Motivation

One-dimensional trend filtering is powerful because its penalty has a clear calculus interpretation. If $f$ is sampled on a path graph, then penalizing second differences encourages piecewise-linear behavior, and penalizing third differences encourages piecewise-quadratic behavior. Algebraically, the unpenalized space is the null space of a finite-difference operator. For example, on a connected path,

$$\mathrm{D}^2 f = 0 \quad \Longleftrightarrow \quad f \text{ is affine along the path,}$$

and

$$\mathrm{D}^3 f = 0 \quad \Longleftrightarrow \quad f \text{ is quadratic along the path.}$$

For a graph that samples a two- or three-dimensional domain, a purely pathwise construction is useful but incomplete. It asks whether $f$ looks polynomial along selected geodesics. A more multivariate construction should ask whether the graph has a meaningful analogue of a gradient, Hessian, and higher derivative tensor.

The first graph derivative is already available: the oriented incidence operator. The new question is how to differentiate it again. That is the central issue of this document.

> *Edge lengths define how far apart adjacent vertices are. A graph Hessian also needs to know which edge directions at neighboring vertices should be compared. That extra datum is a discrete connection or parallel-transport rule.*

## 2 Metric Graphs, Darts, and Directional Fibers

Let

$$G = (\mathcal{V}, \mathcal{E}, \ell)$$

be an undirected graph with positive edge lengths $\ell_{uv} > 0$. Instead of choosing one global orientation per edge, it is convenient to use both directed versions of every edge:

$$\overrightarrow{\mathcal{E}} = \{u \to v : \{u, v\} \in \mathcal{E}\}.$$

We call these directed edges *darts*. For a function

$$f : \mathcal{V} \to \mathbb{R},$$

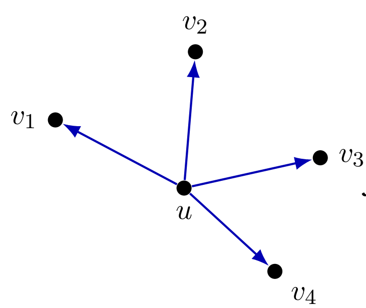
define the length-normalized directional difference along the dart $u \to v$:

$$\delta_{u \to v} f = \frac{f(v) - f(u)}{\ell_{uv}}.$$

The first-difference fiber at $u$ is the vector of outgoing directional differences:

$$\mathcal{F}_u(f) = \big(\delta_{u \to v} f\big)_{v \in \mathcal{N}(u)} \in \mathbb{R}^{\mathcal{N}(u)}.$$

This is a graph analogue of the directional derivative of $f$ in every available outgoing edge direction.




$$\mathcal{F}_u(f) = \left(\frac{f(v_1) - f(u)}{\ell_{uv_1}}, \ldots, \frac{f(v_4) - f(u)}{\ell_{uv_4}}\right)$$

Figure 1: The first-difference fiber at a vertex is the collection of length-normalized outgoing differences. It is the discrete object that plays the role of a local directional gradient.

## 3 Why Raw All-Pairs Fiber Differences Are Too Strong

A tempting second-difference construction is to compare every outgoing directional derivative at $u'$ with every outgoing directional derivative at $u$ across a base dart $u \to u'$. This gives entries of the form

$$\delta_{u' \to v'} f - \delta_{u \to v} f, \qquad v \in \mathcal{N}(u), \quad v' \in \mathcal{N}(u').$$

This is mathematically definable, but it is not the graph Hessian we want. It compares unrelated directional derivatives.

The problem already appears on a square grid. Orient local directions by the coordinate axes and take

$$f(i, j) = i.$$

Then

$$\Delta_x f = 1, \qquad \Delta_y f = 0.$$

An all-pairs comparison includes terms such as

$$\Delta_y f(i + 1, j) - \Delta_x f(i, j) = 0 - 1 = -1.$$

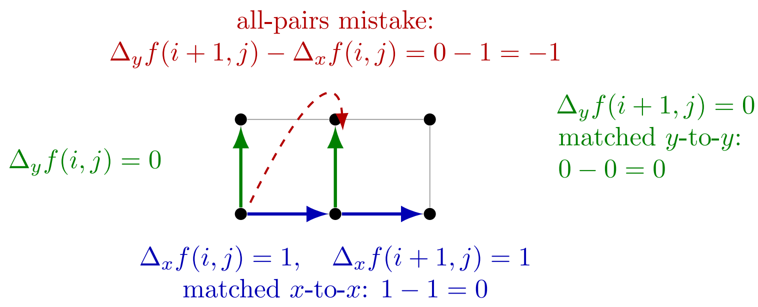



Figure 2: On a grid, all-pairs fiber comparison mixes unrelated directions. For $f(i,j) = i$, comparing a $y$-direction derivative with an $x$-direction derivative produces a nonzero second difference even though $f$ is affine.

Thus a raw all-pairs second-difference penalty would not annihilate an affine function on a grid. A graph Hessian should annihilate affine functions.

This example shows why a second derivative needs a direction-matching rule. In smooth geometry, directional derivatives at neighboring points are compared after parallel transport. The graph version needs the same idea.

## 4 The Abstract Transported Graph Hessian

Let $a = u \to u'$ be a base dart. A direction-transport rule assigns a matrix

$$T_{u\to u'} : \mathbb{R}^{\mathcal{N}(u')} \to \mathbb{R}^{\mathcal{N}(u)}.$$

The entry $T_{u\to u'}(v, v')$ describes how much the outgoing direction $u' \to v'$ contributes to the transported version of the direction $u \to v$.

The transported second difference is

$$\left(\mathrm{D}_T^2 f\right)_{u\to u'}(v) = \sum_{v'\in\mathcal{N}(u')} T_{u\to u'}(v, v')\,\delta_{u'\to v'} f - \delta_{u\to v} f.$$

Equivalently,

$$\mathrm{D}_T^2 f(u \to u') = T_{u\to u'} \mathcal{F}_{u'}(f) - \mathcal{F}_u(f).$$

This is the graph-Hessian object. It compares the directional fiber at $u'$ with the directional fiber at $u$, but only after transporting the $u'$-fiber into the $u$-fiber. The construction is linear in $f$ as long as the transport matrices are built from the graph geometry and held fixed during fitting.

## 5 Direction-Matching and Parallel-Transport Rules

The abstract operator is only useful after specifying $T_{u\to u'}$. This section lists transport rules that are compatible with the graph-trend-filtering project. They range from exact grid references to geometry-estimated soft matching.

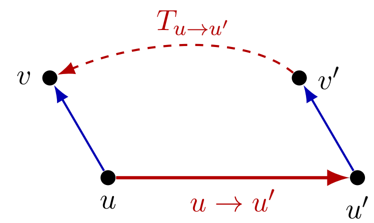



Figure 3: The transported Hessian compares outgoing directional derivatives after a direction at $u'$ has been matched or transported to a direction at $u$. The transport matrix is the extra structure missing from an edge-length only graph.

### 5.1 Exact Coordinate Matching

On a coordinate grid, each dart has a coordinate direction label:

$$+x,\ -x,\ +y,\ -y,\ldots.$$

For a base dart $u \to u'$, define

$$T_{u\to u'}(v,v') = \begin{cases} 1, & \text{if } u \to v \text{ and } u' \to v' \text{ have the same coordinate direction,} \\ 0, & \text{otherwise.} \end{cases}$$

This rule recovers ordinary finite-difference Hessian entries on a lattice. It is the first implementation reference because the expected null spaces are known:

$$\mathrm{D}^1 f = 0 \iff f \text{ is constant}, \qquad \mathrm{D}^2_T f = 0 \iff f \text{ is affine}$$

on a connected rectangular grid, away from boundary conventions.

### 5.2 Edge-Angle Riemannian Graph Matching

Suppose the graph has been enriched with angles between incident edges. For neighbors $v, w \in \mathcal{N}(u)$, let

$$\cos\theta_u(v,w)$$

be the cosine of the angle between the darts $u \to v$ and $u \to w$. Across a base dart $u \to u'$, the direction $u' \to v'$ should match $u \to v$ when both have similar orientation relative to the base edge. Since the base direction reverses when viewed from $u'$, one natural mismatch score is

$$S_{u\to u'}(v,v') = \left|\cos\theta_u(v,u') + \cos\theta_{u'}(v',u)\right|.$$

Small $S$ means that the two directions make compatible angles with the base edge.

A hard matching chooses the best $v'$ for each $v$, but only if the match is good enough. Thus a hard rule should include an acceptance threshold:

$$v^\star(v) = \arg\min_{v'\in\mathcal{N}(u')} S_{u\to u'}(v,v'), \qquad T_{u\to u'}(v,v^\star) = 1 \quad \text{only if} \quad S_{u\to u'}(v,v^\star) \le s_{\max}.$$

If the best score is larger than $s_{\max}$, that direction is left unmatched or handled by a boundary/sparse-neighborhood rule. The threshold $s_{\max}$ is therefore the main modeling choice in hard matching.

A soft matching uses

$$T_{u\to u'}(v,v') = \frac{\exp\{-S_{u\to u'}(v,v')^2/\tau^2\}}{\sum_{w\in\mathcal{N}(u')}\exp\{-S_{u\to u'}(v,w)^2/\tau^2\}}.$$

The bandwidth $\tau$ controls how sharply directions are matched. In practice $\tau$ should be chosen from the empirical distribution of available nonzero mismatch scores. A robust default is

$$\tau = c_\tau \ \mathrm{median}\{S_{u\to u'}(v,v') : \text{candidate pairs in local disks}\},$$

with $c_\tau \in [0.5, 1]$. A more local version uses the median score inside the current disk or for the current base dart. The effective entropy of each row of $T$ should be reported: if all mass falls on one direction, the rule is nearly hard matching; if the mass is almost uniform, $\tau$ is too large.

This rule is close to the earlier idea of enriching a weighted graph into a Riemannian graph: edge lengths give metric scale, and incident-edge angles give local tangent geometry.

### 5.3 Local Embedding Soft Transport

When angles are not given, we can estimate them in local coordinates. For a base dart $u \to u'$, choose a graph disk containing $u$, $u'$, and their neighbors. Embed this disk in $\mathbb{R}^m$ using a local coordinate method such as metric MDS or weighted-GRIP followed by edge-KK refinement. Let $z_w$ denote the local coordinate of vertex $w$.

Define unit edge directions

$$\xi_u(v) = \frac{z_v - z_u}{\|z_v - z_u\|}, \qquad \xi_{u'}(v') = \frac{z_{v'} - z_{u'}}{\|z_{v'} - z_{u'}\|}.$$

If both fibers are represented in the same local chart, then

$$T_{u\to u'}(v,v') = \frac{\exp\{-\left\|\xi_u(v) - \xi_{u'}(v')\right\|^2/(2\sigma_\theta^2)\}}{\sum_{w\in\mathcal{N}(u')}\exp\{-\left\|\xi_u(v) - \xi_{u'}(w)\right\|^2/(2\sigma_\theta^2)\}}.$$

If separate charts are used at $u$ and $u'$, first align the overlapping vertices by an orthogonal Procrustes map. This makes the rule a discrete parallel transport induced by local coordinate frames.

The bandwidth $\sigma_\theta$ is the angular analogue of $\tau$. Since $\xi_u(v)$ and $\xi_{u'}(v')$ are unit vectors,

$$\left\|\xi_u(v) - \xi_{u'}(v')\right\|^2 = 2\{1 - \cos\angle(\xi_u(v), \xi_{u'}(v'))\}.$$

Therefore $\sigma_\theta$ can be specified as an angular bandwidth. If a typical acceptable angular mismatch is $\alpha_0$, set

$$\sigma_\theta^2 = 2(1 - \cos\alpha_0).$$

For example, $\alpha_0 = 20^\circ$ gives a fairly sharp match, while $\alpha_0 = 35^\circ$ is more forgiving. A data-adaptive alternative is to set $\sigma_\theta$ to the median nearest-neighbor directional mismatch among candidate direction pairs in the local disk.

This is the most practical rule for data-derived kNN graphs. It accepts that high-degree neighborhoods contain many sampled directions and uses soft matching rather than forcing one-to-one correspondences.

### 5.4 Adaptive Local Transport Thresholding

The soft transport rule assigns weights to candidate directions, but the operator still needs to decide whether a transported comparison is trustworthy enough to become a Hessian row. A fixed global threshold is often too crude. The same numerical margin can mean different things at a path endpoint, a grid interior vertex, and a high-degree $k$-NN neighborhood. The practical rule should therefore judge a candidate match relative to the local score distribution available at the current transported fiber.

For a base dart $u \to u'$, a source direction $u \to v$, and target candidates $u' \to v'_j$, write

$$s_j = S_{u\to u'}(v, v'_j), \qquad j = 1, \dots, J,$$

and let

$$s_{(1)} \le s_{(2)} \le \cdots \le s_{(J)}$$

be the ordered scores. The best candidate is useful only when it is locally good and locally separated from the alternatives. The best-versus-second margin is

$$m = s_{(2)} - s_{(1)}.$$

**Local quantile rule.** Let $\widehat{F}_s$ be the empirical distribution function of the local candidate scores and let $\widehat{F}_{\Delta s}$ be the empirical distribution function of the adjacent score gaps

$$\Delta s_j = s_{(j+1)} - s_{(j)}, \qquad j = 1, \dots, J-1.$$

Define

$$q_s = \widehat{F}_s(s_{(1)}), \qquad q_m = \widehat{F}_{\Delta s}(m).$$

A local quantile gate accepts the row only if

$$q_s \le q_{\max}, \qquad q_m \ge q_{\min}.$$

The first inequality says that the best score is in the lower tail of the local score distribution. The second says that the best score is separated from the runner-up by a locally meaningful gap.

**Local robust-$z$ rule.** An alternative is to measure the same two ideas using robust local scales. Let

$$\mathrm{scale}(x) = 1.4826\ \mathrm{MAD}(x)$$

with fallback to interquartile range or standard deviation when the median absolute deviation is zero. Define

$$z_s = \frac{\mathrm{median}(s) - s_{(1)}}{\mathrm{scale}(s)}, \qquad z_m = \frac{m - \mathrm{median}(\Delta s)}{\mathrm{scale}(\Delta s)}.$$

A robust-$z$ gate accepts the row only if

$$z_s \ge z_{s,\min}, \qquad z_m \ge z_{m,\min}.$$

Large $z_s$ means the best candidate is unusually good relative to the local candidate cloud. Large $z_m$ means the best candidate is unusually well separated from the next best candidate.

**Entropy-fraction rule.** The softmax weights

$$w_j = \frac{\exp\{-s_j^2/\tau^2\}}{\sum_{k=1}^{J} \exp\{-s_k^2/\tau^2\}}$$

also provide a useful ambiguity diagnostic. Define the softmax entropy

$$H = -\sum_{j=1}^{J} w_j \log w_j,$$

the effective number of matched directions

$$n_{\text{eff}} = \exp(H),$$

and the local effective-match fraction

$$\rho_H = \frac{n_{\text{eff}}}{J}.$$

The fraction $\rho_H$ is close to $1/J$ when the match is concentrated on one direction, and close to 1 when the match is nearly uniform. A row can therefore be rejected when

$$\rho_H > \rho_{\max}.$$

This makes the entropy threshold comparable across vertices of different degree.

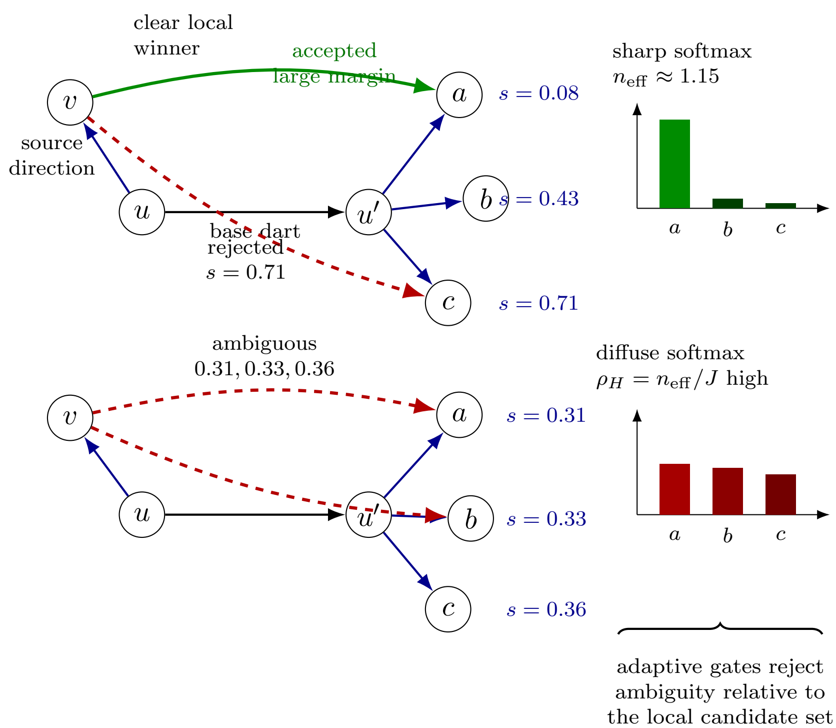



Figure 4: Adaptive local thresholding for transported direction matching. The top neighborhood has one clearly best candidate, a large margin, and a sharp softmax distribution; it is accepted. The bottom neighborhood has several nearly tied candidates and diffuse softmax mass; it is rejected by local quantile, robust-$z$, or entropy-fraction gates.

These thresholds are best understood as diagnostic and experimental controls, not as a final mathematical axiom. They decide which transported directional comparisons are trusted before they enter the Hessian operator. Every accepted or dropped row should carry the corresponding diagnostics: best score, second score, margin, robust-$z$ values, entropy, effective-match fraction, and drop reason. This keeps the operator auditable on irregular graphs.

### 5.5 Status of the Adaptive-Threshold Calibration Experiment

The first calibration sweep treated adaptive thresholding as a retention constrained model-selection problem. For each graph, threshold setting, and transport backend, the experiment measured the linear-polynomial residual

$$R_{\text{linear}} = \frac{\|AP_{\text{linear}}\|_F}{\|P_{\text{linear}}\|_F}$$

subject to the practical constraint

$$\rho_{\text{row}} = \frac{\#\{\text{accepted rows}\}}{\#\{\text{candidate rows}\}} \geq 0.50.$$

This constraint is essential. Very low residuals are not meaningful if they are obtained by dropping most Hessian rows.

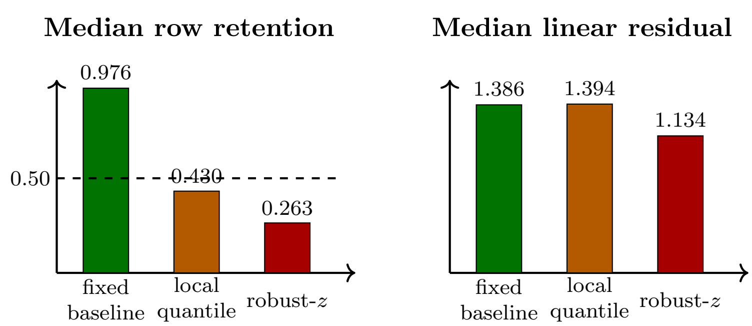



Figure 5: Summary of the adaptive-threshold calibration sweep. Adaptive gates can reduce residuals, but the reduction often comes with substantial row loss. The fixed angle/length rule remains the safest broad baseline under the 50% row-retention constraint.

The sweep tested 294 operator constructions and all runs completed. The main finding is not that adaptive gates are useless; rather, they are not yet a universal default. They won on clean path and grid references, but the fixed angle/length gate won on jittered grids and sampled symmetric $k$-NN graphs. This is exactly the pattern one would expect if local score filters are good calibration diagnostics but can become overly aggressive on irregular real-data graphs.

The conclusion for the next phase is therefore to stop adding more gates to the same soft-matching rule and instead implement a qualitatively different parallel-transport mechanism.

### 5.6 Approximate Parallelogram Transport

There is also a parallelogram idea: match $u \to v$ with $u' \to v'$ when the four vertices approximately close a translated quadrilateral. There are two versions, and they should not be confused.

For a base dart $u \to u'$, define

$$Q^G_{u\to u'}(v, v') = \left|d_G(v, v') - \ell_{uu'}\right| + \left|\ell_{uv} - \ell_{u'v'}\right|.$$

Small $Q^G$ means that $v'$ is close to where $v$ would be translated by the displacement $u \to u'$, using only graph distances. This is an appealing intrinsic formula, but it is also strong. In a generic sparse graph, $d_G(v, v') \approx \ell_{uu'}$ may fail simply because the graph has no short edge or short path between $v$ and $v'$, even when the ambient or latent geometry says the two directions are parallel.

For data-derived kNN graphs with ambient coordinates $x_u$, or with reliable local isometric embedding coordinates $z_u$, the more practical parallelogram score is

$$Q^E_{u\to u'}(v, v') = \big|\|x_v - x_{v'}\| - \|x_u - x_{u'}\|\big| + \big|\|x_u - x_v\| - \|x_{u'} - x_{v'}\|\big| \, .$$

The same formula can be used with $z$-coordinates instead of $x$-coordinates when ambient coordinates are unavailable or when one wants an intrinsic local embedding. A still sharper coordinate version also checks displacement-vector closure:

$$Q^{\text{vec}}_{u\to u'}(v, v') = \|(x_{v'} - x_{u'}) - (x_v - x_u)\| \, .$$

A soft transport can then use whichever score is appropriate:

$$T_{u\to u'}(v, v') = \frac{\exp\{-Q_{u\to u'}(v, v')^2/\tau_Q^2\}}{\sum_{w\in\mathcal{N}(u')} \exp\{-Q_{u\to u'}(v, w)^2/\tau_Q^2\}}.$$

Here $Q$ may be $Q^G$, $Q^E$, or $Q^{\text{vec}}$. In practice, $Q^E$ or $Q^{\text{vec}}$ should be preferred for kNN graphs when ambient coordinates are available.

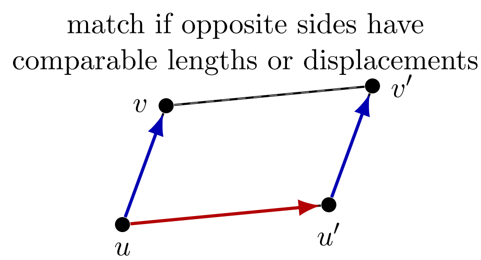



Figure 6: Approximate parallelogram transport matches edge directions by asking whether the four vertices close a parallelogram. On generic sparse graphs, an ambient or local-embedding score is often more practical than a pure graph-distance score.

### 5.7 Regression-Based Gradient Transport

The previous rules match individual edge directions. A different approach is to estimate a local gradient vector at each vertex and then compare gradients.

In local coordinates $z_w$ around $u$, fit

$$f(w) - f(u) \approx g_u^\top (z_w - z_u)$$

by weighted least squares. The fitted gradient is linear in $f$:

$$g_u = L_u f$$

for a matrix $L_u$ determined by the local coordinates and weights.

The coordinates $z_w$ may come from three sources:

(i) ambient coordinates, when the graph was constructed from observed points $x_w \in \mathbb{R}^p$;

(ii) local isometric embedding coordinates from graph distances, such as metric MDS or weighted-GRIP followed by edge-KK refinement;

(iii) coordinates supplied by an enriched Riemannian graph representation.

For synthetic kNN benchmarks, ambient coordinates are useful for debugging. For the intended graph-intrinsic method, the default should be local isometric coordinates from a graph disk.

The disk radius should be chosen adaptively rather than fixed globally. A reasonable rule is

$$r_u = \min\{r : \#D_G(u, r) \geq n_{\min} \text{ and } \#E(D_G(u, r)) \geq e_{\min}\},$$

subject to $r_u \leq r_{\max}$. The minimum vertex count should exceed the number of regression coefficients. For an $m$-dimensional local linear fit, one needs at least $m + 1$ nondegenerate points, but in practice

$$n_{\min} \approx 3(m + 1) \quad \text{to} \quad 5(m + 1)$$

is safer. The chosen disk should also pass a conditioning check on the local design matrix. If the local design is ill-conditioned, the algorithm should increase the radius, reduce the estimated local dimension, or mark the gradient as unreliable.

Across a base dart $u \to u'$, align the coordinate frames by a map $\Pi_{u' \to u}$, and define

$$\mathcal{H}_{u \to u'} f = \frac{\Pi_{u' \to u} g_{u'} - g_u}{\ell_{uu'}}.$$

This is a vector-valued second difference. It bypasses the variable degree of graph vertices and may be the numerically cleanest way to build Hessian-like rows on high-degree kNN graphs.

The tradeoff is conceptual: this rule is less purely incidence-based. It is a local regression derivative, not a direct comparison of raw edge differences.

# 6 Trend-Filtering Penalties

Once $T$ is fixed, the transported Hessian is a linear operator in $f$. The order-two transported graph trend-filtering estimator is

$$\widehat{f}_\lambda = \arg\min_{f \in \mathbb{R}^{|\mathcal{V}|}} \frac{1}{2} \sum_{u \in \mathcal{V}} (y_u - f_u)^2 + \lambda \left\| \mathrm{D}^2_T f \right\|_1.$$

This is the $\ell_1$-adaptive analogue of a graph Hessian smoother. The $\ell_1$ penalty encourages many transported directional-gradient changes to be exactly zero, so the fitted signal is locally affine except across selected graph regions where the Hessian-like quantity is allowed to jump.

If $D^1$ denotes the length-normalized dart incidence operator, then the first-order case is graph total variation:

$$\widehat{f}_\lambda = \arg\min_{f} \frac{1}{2} \|y - f\|_2^2 + \lambda \left\| D^1 f \right\|_1.$$

The transported second-order case is the candidate graph analogue of linear trend filtering on manifolds or metric graphs.

# 7 Higher Differences

The same idea can be extended recursively, but the notation should be handled carefully. Let $\mathcal{G}_1(u; f) = \mathcal{F}_u(f)$. Suppose $\mathcal{G}_r(u; f)$ is a vector of order-$r$ directional quantities based at $u$. A higher-order transport rule assigns

$$T^{(r)}_{u\to u'} : \mathcal{G}_r(u'; f) \to \mathcal{G}_r(u; f).$$

Then

$$\mathcal{G}_{r+1}(u \to u'; f) = T^{(r)}_{u\to u'} \mathcal{G}_r(u'; f) - \mathcal{G}_r(u; f).$$

For $r = 1$, this gives the transported Hessian. For $r = 2$, it compares Hessian-like fibers and gives a third-difference operator. The corresponding penalty

$$\left\|\mathrm{D}^3_T f\right\|_1$$

should encourage piecewise-quadratic behavior when the transport rule is consistent with the underlying geometry.

On a path graph, each relevant fiber is one-dimensional and transport is trivial. The recursion collapses to ordinary one-dimensional finite differences:

$$(\mathrm{D}^r f)_i = \sum_{q=0}^{r} (-1)^{r-q} \binom{r}{q} f_{i+q}.$$

# 8 Regression-Based Order-3 Operators

The experiments with edge-angle matching are useful because they make a negative result visible. Edge-angle transport is a geometrically natural baseline, but on generic data-derived $k$-NN graphs sampled from quadric surfaces it does not preserve intrinsic affine probes nearly as well as regression-gradient transport. This suggests that the next order-3 operator should not be an order-3 extension of edge-angle matching. The more promising route is to extend the successful regression-based idea.

## 8.1 Recap: Order-2 Regression-Gradient Transport

The order-2 regression-gradient construction fits, at each vertex $u$, a local linear model

$$f(w) - f(u) \approx g_u^\top (z_w - z_u), \qquad w \in D_u,$$

where $D_u$ is a local graph disk and $z_w$ are coordinates for the vertices in that disk. The fitted gradient is linear in the sampled values of the function:

$$g_u = L_u f.$$

When $z_w$ are supplied global coordinates, such as the data matrix used to construct a $k$-NN graph, the gradient coefficients $g_u$ and $g_v$ at neighboring vertices are expressed in the same coordinate basis. A transported second-difference row can therefore compare them directly:

$$(A_2 f)_{u\to v,a} = (g_v)_a - (g_u)_a.$$

If desired, this can be divided by the base-edge length $\ell_{uv}$ to put the row on a derivative scale:

$$(A_2 f)_{u\to v,a} = \frac{(g_v)_a - (g_u)_a}{\ell_{uv}}.$$

The essential null-space behavior is

$$A_2 p \approx 0 \quad \text{for coordinate polynomials } p \text{ of degree } \leq 1.$$

Thus affine probes should have small residual, while quadratic probes should generally be detected.

### 8.2 Local Quadratic Regression for Order 3

The natural order-3 analogue is to estimate a local quadratic Taylor model instead of only a local linear model. Around each vertex $u$, fit

$$f(w) - f(u) \approx a_u + g_u^\top (z_w - z_u) + \frac{1}{2}(z_w - z_u)^\top H_u (z_w - z_u), \qquad w \in D_u.$$

The intercept $a_u$ can be omitted if the model is centered so that the left-hand side is $f(w) - f(u)$, but keeping it in the displayed formula makes the analogy with local polynomial regression explicit. The Hessian coefficients $H_u$ are again linear in $f$:

$$\operatorname{vech}(H_u) = Q_u f,$$

where vech stores the unique symmetric entries of $H_u$. If the coordinate dimension is $m$, there are

$$\frac{m(m+1)}{2}$$

such second-derivative coefficients.

For a base dart $u \to v$, define the supplied-coordinate order-3 rows by comparing the local Hessian coefficients:

$$(A_3 f)_{u \to v, ab} = \frac{(H_v)_{ab} - (H_u)_{ab}}{\ell_{uv}}, \qquad 1 \leq a \leq b \leq m.$$

This is the coordinate-regression analogue of a third derivative. Its target null-space behavior is

$$A_3 p \approx 0 \quad \text{for coordinate polynomials } p \text{ of degree } \leq 2.$$

Consequently the expected diagnostic pattern is

$$R_{\text{aff}} \approx 0, \qquad R_{\text{quad}} \approx 0, \qquad R_{\text{cubic}} > 0.$$

Here

$$R_{\mathcal{P}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \frac{\|A_3 p\|_2}{\|p\|_2}$$

is the mean residual over a family of polynomial probes.

### 8.3 Why Supplied Coordinates Should Come First

The first implementation should use supplied coordinates. This includes the intrinsic coordinates in synthetic oracle tests and, more importantly, the ambient data coordinates available for data-derived $k$-NN graphs. In this setting every local fit uses one shared coordinate basis, so the comparison

$$H_v - H_u$$

is meaningful without an additional tensor-transport rule.

Graph-derived local charts are conceptually different. If $H_u$ and $H_v$ are estimated in independently embedded local disks, their coordinate bases need not agree. A Hessian tensor at $v$ must first be transported into the chart at $u$:

$$H_v \longmapsto R_{u\leftarrow v}^{\top} H_v R_{u\leftarrow v},$$

where $R_{u\leftarrow v}$ is an estimated chart-to-chart alignment or parallel-transport map. This is a real geometric design problem. It should be deferred until the supplied-coordinate order-3 operator has passed the basic null-space diagnostics.

### 8.4 Recommended Phase for the Next Landing Patch

The next implementation phase should therefore be:

(i) build a local quadratic-regression coefficient map $Q_u f = \mathrm{vech}(H_u)$ for supplied coordinates;

(ii) assemble $A_3$ by comparing $Q_v f - Q_u f$ across graph darts;

(iii) validate the operator on path graphs, rectangular grids, sampled two-dimensional quadrics, and sampled three-dimensional quadrics;

(iv) compare intrinsic-oracle coordinates with ambient data coordinates;

(v) report affine, quadratic, cubic, and fractional-power probe residuals.

| Test setting | Expected behavior |
|---|---|
| Path graph | Agreement with ordinary one-dimensional third differences, up to boundary conventions. |
| Rectangular grid | Coordinate quadratics should be annihilated by the order-3 operator. |
| Sampled 2D quadrics | Intrinsic coordinates are the oracle; ambient data coordinates should be close when curvature and sampling are moderate. |
| Sampled 3D quadrics | The same intrinsic-versus-ambient test should hold in higher intrinsic dimension. |
| Cubic probes | Residuals should be nonzero, confirming that the operator is not too weak. |
| Fractional-power probes | These are diagnostic response curves, not null-space claims. |

This phase keeps the implementation mathematically narrow. It uses the strongest empirical signal so far: regression-based transported derivatives preserve the intended polynomial null space much better than direction matching alone on irregular data-derived graphs.

## 9 Null-Space Questions

The central validation question is not only whether the operator can be built, but whether its null space has the intended interpretation.

### 9.1 First Order

For a connected graph,

$$D^1 f = 0 \quad \Longleftrightarrow \quad f \text{ is constant.}$$

This is independent of transport.

### 9.2 Second Order

For a good transported Hessian,

$$\mathrm{D}_T^2 f = 0$$

should mean that $f$ is locally affine in the graph geometry. On a coordinate grid this should exactly recover affine coordinate functions:

$$f(i,j) = a + bi + cj.$$

On a sampled manifold graph, it should approximately annihilate affine functions in local intrinsic coordinates.

### 9.3 Third Order

For a good transported third difference,

$$\mathrm{D}_T^3 f = 0$$

should mean that $f$ is locally quadratic. On grids, this means ordinary quadratic coordinate polynomials. On sampled surfaces, the test should be intrinsic rather than ambient whenever possible.

These expectations should be treated as empirical requirements. Candidate transport rules should be compared by

$$\dim\ker(\mathrm{D}_T^r) \qquad \text{and} \qquad \frac{\|\mathrm{D}_T^r P_d\|_F}{\|P_d\|_F},$$

where $P_d$ is a matrix of known polynomial probe functions.

## 10 Recommended Implementation Roadmap

The implementation should be staged so that each mathematical assumption is testable before it is used inside a penalized regression fit. In particular, the first deliverable should not be a new smoother. It should be an operator constructor that makes the transport rule visible.

### 10.1 Phase 0: Common Operator Interface

The first step is to define one common interface for every transport rule. For each base dart $a = u \to u'$, the transport constructor should return

$$T_{u\to u'} : \mathbb{R}^{\mathcal{N}(u')} \to \mathbb{R}^{\mathcal{N}(u)}.$$

The downstream Hessian assembly should not need to know whether $T$ came from exact grid labels, local embeddings, edge-angle data, or a hard matching rule. It should always form rows of the same type:

$$\left(\mathrm{D}_T^2 f\right)_{u\to u'}(v) = \sum_{v'\in\mathcal{N}(u')} T_{u\to u'}(v,v')\delta_{u'\to v'} f - \delta_{u\to v} f.$$

The phase output should include the sparse operator $A_T$, the list of base darts, the local direction labels $v \in \mathcal{N}(u)$, and enough metadata to trace each matrix row back to a geometric comparison.

### 10.2 Phase 1: Exact Reference Transports

The first implemented transport rules should be exact references on graphs where the correct answer is known.

- **Path graph.** Directions are the two orientations along the path. The operator should reduce to ordinary one-dimensional finite differences away from boundaries.
- **Square grid.** Directions are coordinate directions. An $x$-direction at $u$ should match the $x$-direction at $u'$, and similarly for $y$. This phase should verify that affine functions $f(i,j) = a+bi+cj$ are annihilated by $\mathrm{D}_T^2$.
- **Cube or lattice graph.** If three-dimensional examples are available, repeat the same test for $x, y, z$ coordinate directions.

This phase is the guardrail against a tempting but wrong all-pairs construction. The acceptance criterion is simple: constants must lie in the null space of $D^1$, affine coordinate functions must have nearly zero $\mathrm{D}_T^2$ residual, and quadratic coordinate functions must have nearly zero $\mathrm{D}_T^3$ residual where complete stencils are available.

### 10.3 Phase 2: Operator Diagnostics Without Fitting

The second phase should expose the operator as a diagnostic object. This should be analogous in spirit to an operator-construction helper: it builds the matrix and returns summaries, but does not fit a response.

The diagnostic object should report:

$$\dim\ker(A_T), \qquad \frac{\|A_T P_d\|_F}{\|P_d\|_F},$$

where $P_d$ is a matrix of polynomial probe functions. It should also return row counts, dropped-row counts, base-dart counts, transport-matrix entropy, match-score distributions, edge-length summaries, boundary/sparse neighborhood omissions, and row-normalization choices.

The most important design principle is that failed transport should be visible. If a candidate direction has no plausible match, the implementation should record whether the row was dropped, mirrored, softly averaged, or kept as a low-confidence comparison.

### 10.4 Phase 3: Local Embedding Soft Transport

After the exact references are stable, implement the first general metric-graph transport using local coordinates. The coordinates $z_w$ may come from ambient coordinates when available, weighted-GRIP $\to$ edge-KK local embeddings, or metric MDS fallback. For a base dart $u \to u'$, candidate directions $u \to v$ and $u' \to v'$ can be scored by

$$S_{u\to u'}(v, v')^2 = \frac{\theta(v, v')^2}{\sigma_\theta^2} + \frac{(\ell_{uv} - \ell_{u'v'})^2}{\sigma_\ell^2},$$

where $\theta(v, v')$ is the angle between the local direction vectors after aligning the coordinate systems. The soft transport is then

$$T_{u \to u'}(v, v') = \frac{\exp\{-S_{u \to u'}(v, v')^2 / \tau^2\}}{\sum_{w \in \mathcal{N}(u')} \exp\{-S_{u \to u'}(v, w)^2 / \tau^2\}}.$$

The bandwidths $\sigma_\theta$, $\sigma_\ell$, and $\tau$ should begin as robust local scales, for example median absolute deviations or median nearest candidate scores inside a disk. This makes the rule adaptive to degree, sampling density, and local anisotropy. The phase should report sensitivity to these scale choices rather than hiding them inside a default.

## 10.5 Phase 4: Hard Matching and Thresholded Transport

Soft transport is stable but can blur directions on high-degree graphs. The next rule should therefore be a hard or thresholded version. For each $u \to v$, choose

$$v_* = \arg\min_{v' \in \mathcal{N}(u')} S_{u \to u'}(v, v').$$

Accept the match only if

$$S_{u \to u'}(v, v_*) \leq q_{u \to u'},$$

where $q_{u \to u'}$ is a fixed or locally estimated threshold. If the best match is too poor, the corresponding row should be dropped or flagged. The main diagnostic is the tradeoff between interpretability and sparsity: tighter thresholds improve match quality but may make the operator too sparse near boundaries or in irregular regions.

The first practical version of this phase should include the adaptive local rules from Figure 4: local score quantiles, local robust-$z$ separation, and effective-match fraction. These rules preserve the hard-thresholded spirit while avoiding a single global cutoff that behaves differently as vertex degree and local sampling geometry change.

## 10.6 Phase 5: Comparative Operator Experiments

The first comparison should be operator-only. The same graphs and polynomial probes should be run through exact grid transport, soft local-embedding transport, hard thresholded transport, and any approximate parallelogram rule. The report should compare:

- nullity versus the expected polynomial dimension;
- polynomial residuals $\|A_T P_d\|_F \,/\, \|P_d\|_F$;
- row count and dropped-row rate;
- transport entropy and match-score distributions;
- sensitivity to disk radius, local embedding dimension, and bandwidth;
- boundary behavior and sparse one-sided neighborhood behavior.

The test graphs should include paths, square grids, nonuniform grids, sampled $[0, 1]^2$, Delaunay one-skeletons, symmetric $k$-NN graphs, and sampled quadric surfaces. The quadric examples are especially important because they separate intrinsic polynomial behavior from ambient-coordinate artifacts.

### 10.7 Phase 6: Penalized Trend-Filtering Fits

Only after the operator diagnostics are convincing should the operator be used inside a trend-filtering estimator:

$$\hat{f}_\lambda = \arg\min_f \frac{1}{2} \|y - f\|_2^2 + \lambda \|A_T f\|_1 .$$

The first landing fit should use fixed $\lambda$ values on small examples so that the behavior of $A_T$ remains visible. Cross-validation, GCV-like criteria, and benchmark-scale comparisons should come after the operator and the fixed-$\lambda$ fit are stable.

This phase should not replace the current pathwise graph trend-filtering experiments. The transported-Hessian operator and the pathwise annihilator operator should be treated as complementary graph trend-filtering families until their null spaces, residual diagnostics, and prediction behavior have been compared systematically.

## 11 Open Design Questions

- **Hard versus soft transport.** Hard matching gives sparse, interpretable rows. Soft matching is more stable on high-degree noisy graphs but creates denser operator rows.
- **Normalization.** First differences should usually be divided by edge length. Second differences may need an additional division by the base-edge length $\ell_{uu'}$ if the goal is derivative-scale rather than difference-scale Hessian entries.
- **Boundary handling.** Boundary vertices have missing directions. Options include dropping incomplete comparisons, one-sided stencils, mirror extensions, or local regression gradient transport.
- **Orientation.** Darts avoid arbitrary global orientation for the first derivative, but higher-order rows still depend on which base darts are included. Symmetric inclusion of both darts may be preferable for invariance.
- **Local dimension.** For local embedding transport, the selected dimension $m_v$ affects direction matching. The local dimension diagnostic machinery already developed for geodesic annihilators should be reused.
- **Relation to pathwise annihilators.** Pathwise rows are robust and exactly recover one-dimensional trend filtering on paths. Transported Hessian rows are more multivariate. Both should remain experimental operator families until their null spaces and prediction behavior are compared systematically.

## 12 Summary

The transported graph Hessian formulation separates the problem into two cleanly testable parts. The first part is the incidence-style directional fiber

$$\mathcal{F}_u(f) = \left( \frac{f(v) - f(u)}{\ell_{uv}} \right)_{v \in \mathcal{N}(u)} .$$

The second part is a graph connection

$$T_{u \to u'} : \mathbb{R}^{\mathcal{N}(u')} \to \mathbb{R}^{\mathcal{N}(u)}$$

that says how directions at neighboring vertices should be compared. Together they define

$$\mathrm{D}^2_T f(u \to u') = T_{u\to u'}\mathcal{F}_{u'}(f) - \mathcal{F}_u(f).$$

This is a principled graph analogue of differentiating the gradient after parallel transport. It recovers standard Hessian finite differences on coordinate grids, and it suggests several practical transport rules for general metric graphs. The next step is not to fit data immediately, but to build operator diagnostics and ask whether the null spaces behave as constants, affine functions, and quadratics in the test geometries where those answers are known.